

HW7: Time-Series Regression for TRK-50 Bike Sales

ISYE 4031 - Summer 2026 - Problems Only Version

Submission Instructions

Include your full name, Georgia Tech email, and GTID in your submitted work. You may complete the homework using JupyterLite, a local Python or R environment, spreadsheet software, handwritten work where appropriate, or another approved environment. Your submission should clearly answer each question and show enough reasoning, formulas, code output, or supporting work to be graded.

Required Data Files

- `trk50_bike_sales.csv`

The quarterly sales of the TRK-50 mountain bike for the previous four years are provided in `trk50_bike_sales.csv`.

The columns are:

- Year: year number 1 through 4;
- Quarter: quarter number 1 through 4;
- Time: sequential time index 1 through 16;
- Sales: quarterly bike sales.

Problem 1(a): Plot the Time Series

Create a time-series plot of quarterly sales against Time.

Problem 1(b): Trend

What kind of trend appears to exist?

Problem 1(c): Seasonal Variation

What type of seasonal variation appears to exist: constant/additive or increasing/multiplicative? Explain using the plot.

Problem 1(d): Transformation Decision

Is a transformation needed to obtain a series that displays constant variation? Explain.

Problem 1(e): Fit a Trend and Seasonal Model

Fit an appropriate regression model with a linear trend and seasonal variation terms. Use Q1 as the baseline and create dummy variables Q2, Q3, and Q4.

Problem 1(f): Statistical Importance of Predictors

Do all independent variables seem to be important? Justify your answer statistically.

Problem 1(g): Define Dummy Variables

Write out the definitions of Q2, Q3, and Q4. Also explain what quarter is represented when $Q2 = Q3 = Q4 = 0$.

Problem 1(h): Forecast Year 5

Predict bike sales in year 5, quarters 1 through 4. Report point estimates and 95% confidence and prediction intervals.

Problem 1(i): Prediction Equation and Manual Calculation

Write the fitted prediction equation and use it to calculate the forecasts for time 17 and time 18 manually.

AI Usage Report

Your submission must include an AI Usage Report following the course AI policy. If you used AI tools, include the tool(s), what you asked them to help with, what output you accepted, changed, or rejected, and how you verified the final work. If prompts or screenshots materially affected your submission, include copied prompt text or screenshots in the report when appropriate. If no AI tools were used, include this statement: "No AI tools were used for this submission. You may include this report at the end of your submission or attach it as a separate file.